

Scheduling in the World of Sports

Frits Spieksma (Eindhoven University of Technology),

joint work with Sten Wessel and Cor Hurkens



Variety

There is an incredible variety in the world of sports.



Variety

There is an incredible variety in the world of sports.



However, one commonality is that they have to meet in time and place

Variety

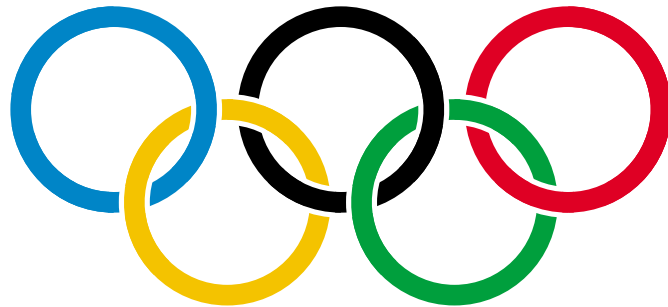
There is an incredible variety in the world of sports.



However, one commonality is that competitors have to meet in time and place: a **schedule** is needed.

This is not an overview

Sports are actually a very old human activity



776 BC

Science related to sports is also old

This is not an overview

Science related to sports also has a long tradition.

- ❑ Kirkman (1847) describes a way to construct a round robin tournament.
 - ❑ Lucas (1883) attributes the “circle method” to Walecki, a French mathematician.
 - ❑ Landau (1895) describes a way to determine the quality of a chess player from the strengths of the players beaten.
-
- ❑ T. P. Kirkman (1847), *On a problem in combinations*, Cam. Dub. Math. J., vol. 2, pp. 191–204.
 - ❑ Edouard Lucas (1883) Édouard Lucas, *Récréations mathématiques*. Paris, France: Gauthier-Villars, 1883, ch. Les jeux de demoiselles, pp. 176–180.
 - ❑ Edmund Landau (1895), *Zur relativen Wertbemessung der Turnierresultate*, Deutsches Wochenschach, 11. Jahrgang, No. 42:51–54, 1895.

Goal

To give a flavor of the variety of scheduling problems by diving into two scheduling problems motivated by fairness challenges in sports.

Contents

□ Ranking Fairness

- A motivating story in chess
- The resulting scheduling problem
- Results (and an open question)

□ Fairness in Incomplete Round Robin Tournaments

- A motivating story in football
- The resulting scheduling problem
- Results

□ Conclusion

Ranking Fairness

A motivating story in chess

□ Once upon a time ...

A motivating story in chess

□ When? Where?

□ January 2002

□ Wijk aan Zee



A motivating story in chess

- ❑ Thus, in January 2002, in Wijk aan Zee, the best chess players in the world gathered to play the Corus Tournament.
- ❑ There are 14 players, the format is a Single Round Robin (SRR), meaning every pair of players meets once.

A motivating story in chess

<u>The players:</u>	<u>Their Elo rating:</u>
Morozevich, Alexander	(2742)
Adams, Michael	(2742)
Leko, Peter	(2713)
Bareev, Evgeny	(2707)
Gelfand, Boris	(2703)
Van Wely, Loek	(2697)
Kasimdzhanov, Rustam	(2695)
Khalifman, Alexander	(2688)
Lautier, Joel	(2687)
Dreev, Alexey	(2683)
Grischuk, Alexander	(2671)
Piket, Jeroen	(2659)
Gurevich, Mikhail	(2641)
Timman, Jan	(2605)

A motivating story in chess

About chess: there is black and there is white

A motivating story in chess

Round 1

Bareev, Evgeny	-Adams, Michael	1/2-1/2
Dreev, Alexey	-Khalifman, Alexander	1/2-1/2
Gelfand, Boris	-Lautier, Joel	1-0
Grischuk, Alexander	-Kasimdzhanov, Rustam	1-0
Gurevich, Mikhail	-Morozovich, Alexander	1/2-1/2
Leko, Peter	-Van Wely, Loek	1-0
Timman, Jan H	-Piket, Jeroen	1/2-1/2

Round 2

Adams, Michael	-Grischuk, Alexander	1-0
Gurevich, Mikhail	-Leko, Peter	1/2-1/2
Kasimdzhanov, Rustam	-Gelfand, Boris	1/2-1/2
Lautier, Joel	-Dreev, Alexey	1/2-1/2
Morozovich, Alexander	-Khalifman, Alexander	1/2-1/2
Piket, Jeroen	-Bareev, Evgeny	1/2-1/2
Van Wely, Loek	-Timman, Jan H	0-1

Round 3

Bareev, Evgeny	-Van Wely, Loek	1-0
Dreev, Alexey	-Kasimdzhanov, Rustam	0-1
Gelfand, Boris	-Adams, Michael	0-1
Grischuk, Alexander	-Piket, Jeroen	1-0
Khalifman, Alexander	-Lautier, Joel	1/2-1/2
Leko, Peter	-Morozovich, Alexander	1/2-1/2
Timman, Jan H	-Gurevich, Mikhail	1/2-1/2

Round 4

Adams, Michael	-Dreev, Alexey	1/2-1/2
Gurevich, Mikhail	-Bareev, Evgeny	1/2-1/2
Kasimdzhanov, Rustam	-Khalifman, Alexander	0-1
Leko, Peter	-Timman, Jan H	1/2-1/2
Morozovich, Alexander	-Lautier, Joel	1-0
Piket, Jeroen	-Gelfand, Boris	1-0
Van Wely, Loek	-Grischuk, Alexander	0-1

Round 5

Bareev, Evgeny	-Leko, Peter	1-0
Dreev, Alexey	-Piket, Jeroen	1-0
Gelfand, Boris	-Van Wely, Loek	1/2-1/2
Grischuk, Alexander	-Gurevich, Mikhail	1/2-1/2
Khalifman, Alexander	-Adams, Michael	1/2-1/2
Lautier, Joel	-Kasimdzhanov, Rustam	1-0
Timman, Jan H	-Morozovich, Alexander	1/2-1/2

Round 6

Adams, Michael	-Lautier, Joel	1/2-1/2
Gurevich, Mikhail	-Gelfand, Boris	0-1
Leko, Peter	-Grischuk, Alexander	1-0
Morozovich, Alexander	-Kasimdzhanov, Rustam	1-0
Piket, Jeroen	-Khalifman, Alexander	1/2-1/2
Timman, Jan H	-Bareev, Evgeny	0-1
Van Wely, Loek	-Dreev, Alexey	1/2-1/2

Round 7

Bareev, Evgeny	-Morozovich, Alexander	1/2-1/2
Dreev, Alexey	-Gurevich, Mikhail	1/2-1/2
Gelfand, Boris	-Leko, Peter	1/2-1/2
Grischuk, Alexander	-Timman, Jan H	1-0
Kasimdzhanov, Rustam	-Adams, Michael	1/2-1/2
Khalifman, Alexander	-Van Wely, Loek	1-0
Lautier, Joel	-Piket, Jeroen	1/2-1/2

Round 8

Bareev, Evgeny	-Grischuk, Alexander	1/2-1/2
Gurevich, Mikhail	-Khalifman, Alexander	1/2-1/2
Leko, Peter	-Dreev, Alexey	1/2-1/2
Morozovich, Alexander	-Adams, Michael	1/2-1/2
Piket, Jeroen	-Kasimdzhanov, Rustam	1/2-1/2
Timman, Jan H	-Gelfand, Boris	1-0
Van Wely, Loek	-Lautier, Joel	1/2-1/2

Round 9

Adams, Michael	-Piket, Jeroen	1/2-1/2
Dreev, Alexey	-Timman, Jan H	1/2-1/2
Gelfand, Boris	-Bareev, Evgeny	1/2-1/2
Grischuk, Alexander	-Morozovich, Alexander	1-0
Kasimdzhanov, Rustam	-Van Wely, Loek	1-0
Khalifman, Alexander	-Leko, Peter	1/2-1/2
Lautier, Joel	-Gurevich, Mikhail	1/2-1/2

Round 10

Bareev, Evgeny	-Dreev, Alexey	1-0
Grischuk, Alexander	-Gelfand, Boris	1/2-1/2
Gurevich, Mikhail	-Kasimdzhanov, Rustam	1-0
Leko, Peter	-Lautier, Joel	1/2-1/2
Morozovich, Alexander	-Piket, Jeroen	1-0
Timman, Jan H	-Khalifman, Alexander	1/2-1/2
Van Wely, Loek	-Adams, Michael	1/2-1/2

Round 11

Adams, Michael	-Gurevich, Mikhail	1-0
Dreev, Alexey	-Grischuk, Alexander	1/2-1/2
Gelfand, Boris	-Morozovich, Alexander	1/2-1/2
Kasimdzhanov, Rustam	-Leko, Peter	1/2-1/2
Khalifman, Alexander	-Bareev, Evgeny	1-0
Lautier, Joel	-Timman, Jan H	1-0
Piket, Jeroen	-Van Wely, Loek	1/2-1/2

Round 12

Bareev, Evgeny	-Lautier, Joel	1-0
Gelfand, Boris	-Dreev, Alexey	1/2-1/2
Grischuk, Alexander	-Khalifman, Alexander	1-0
Gurevich, Mikhail	-Piket, Jeroen	0-1
Leko, Peter	-Adams, Michael	1/2-1/2
Morozovich, Alexander	-Van Wely, Loek	1-0
Timman, Jan H	-Kasimdzhanov, Rustam	1/2-1/2

Round 13

Adams, Michael	-Timman, Jan H	1/2-1/2
Dreev, Alexey	-Morozovich, Alexander	1/2-1/2
Kasimdzhanov, Rustam	-Bareev, Evgeny	0-1
Khalifman, Alexander	-Gelfand, Boris	1/2-1/2
Lautier, Joel	-Grischuk, Alexander	1/2-1/2
Piket, Jeroen	-Leko, Peter	1/2-1/2
Van Wely, Loek	-Gurevich, Mikhail	1/2-1/2

A motivating story in chess

	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13
1 Morozevich													
2 Adams													
3 Leko													
4 Bareev													
5 Gelfand													
6 Van Wely													
7 Kasimdzhanov													
8 Khalifman													
9 Lautier													
10 Dreev													
11 Grischuk													
12 Piket													
13 Gurevich													
14 Timman													

The schedule, view 1: black and white over the rounds

A motivating story in chess

	R1	R2	R3	R4	R5	R6	R7	R8	R9	R10	R11	R12	R13
1 Morozevich	■		■		■		■		■		■		■
2 Adams	■		■		■		■	■		■		■	
3 Leko		■			■		■		■		■		■
4 Bareev		■		■		■			■		■		■
5 Gelfand		■		■		■		■		■			■
6 Van Wely	■		■		■		■		■		■	■	
7 Kasimdzhanov	■		■		■	■		■		■		■	
8 Khalifman	■	■		■		■		■		■		■	
9 Lautier	■		■	■		■		■		■		■	
10 Dreev		■		■		■		■		■		■	
11 Grischuk		■		■		■		■			■		■
12 Piket	■		■		■		■		■	■		■	
13 Gurevich			■		■		■		■		■		■
14 Timman		■		■			■		■		■		■

A few comments: (i) some play 6 matches with white, others 7, (ii) there is a clear alternating pattern for each of the players, (iii) Khalifman plays with black in the first two away matches, while Gurevich has white. All in all, this seems a fine schedule ... but is it?

A motivating story in chess

1 Morozevich	2	3	4	5	6	7	8	9	10	11	12	13	14
2 Adams	1	3	4	5	6	7	8	9	10	11	12	13	14
3 Leko	1	2	4	5	6	7	8	9	10	11	12	13	14
4 Bareev	1	2	3	5	6	7	8	9	10	11	12	13	14
5 Gelfand	1	2	3	4	6	7	8	9	10	11	12	13	14
6 Van Wely	1	2	3	4	5	7	8	9	10	11	12	13	14
7 Kasimdzhanov	1	2	3	4	5	6	8	9	10	11	12	13	14
8 Khalifman	1	2	3	4	5	6	7	9	10	11	12	13	14
9 Lautier	1	2	3	4	5	6	7	8	10	11	12	13	14
10 Dreev	1	2	3	4	5	6	7	8	9	11	12	13	14
11 Grischuk	1	2	3	4	5	6	7	8	9	10	12	13	14
12 Piket	1	2	3	4	5	6	7	8	9	10	11	13	14
13 Gurevich	1	2	3	4	5	6	7	8	9	10	11	12	14
14 Timman	1	2	3	4	5	6	7	8	9	10	11	12	13

The schedule, view 2: black and white over the players ranked by strength

A motivating story in chess

THE ISSUE:

- ❑ In a Single Round Robin where players/teams can be ranked by strength, and where there is an asymmetry in a single match (white/black or home/away), the asymmetries should be evenly distributed over the ranked opponents.
- ❑ This leads us to the concept of Ranking Fairness

An observation

- Playing white in chess is equivalent to playing home.
- Thus, this phenomenon is relevant for each SRR where the teams have home venues.
 - Dutch Baseball (a triple Round Robin (3RR))
 - Danish football (when it was a 3RR)
 - Argentina Football
 -

The setting

- We deal with a Single Round Robin, i.e., each pair of players (or teams) meets once.
- We let n denote the number of players, n is even.
- There are $n-1$ rounds, each round consisting of $n/2$ matches such that each player plays one match in each round.
- The players are ranked with respect to their strength. In other words, a *ranking* is given.

Terminology

- Given a schedule, the sequence of home (H) and away (A) matches a player plays is called its *Home Away Pattern (HAP)*.
- If a player has two consecutive home games, or two consecutive away games, say in rounds r and $r+1$, we say that the player has a *break* in round $r+1$, $r = 1, \dots, n-1$; notice we compute modulo $n-1$.
- If a player has one break, its HAP is called *single break*; if all HAPs in the HAP-set are single break, the HAP-set is called a *single break HAP-set*. We restrict our attention to single break HAP-sets.

Facts

- ❑ FACT: All single break HAP-sets are complementary
- ❑ FACT: All single break HAP-sets have breaks in $n/2$ different rounds
- ❑ We can characterize any single break HAP-set H by specifying the rounds in $\{r_1, \dots, r_{n/2}\}$ in which the breaks occur.
- ❑ We refer to the differences of consecutive rounds with breaks as a D-sequence.
- ❑ A popular D-sequence is the Canonical Pattern Set with D-sequence 222...21

A new concept: ranking fairness

Consider the *ranking HAP*, where the home and away matches of a team p are ordered according to the rank of its opponents $1, \dots, p-1, p+1, \dots, n$. Ideally, the ranking HAP is perfectly alternating (either HAHAAH... or AHAHA...). We call a schedule with this property *ranking fair*.

Thus, a schedule with the following ranking HAP is called ranking fair

	Team 1	Team 2	Team 3		Team n
Team 1	/	H	A		H
Team 2	A	/	H		A
Team 3	H	A	/		H
				/	
Team n	A	H	A		/

A new concept: ranking fairness

	Team 1	Team 2	Team 3		Team n
Team 1	/	H	A		H
Team 2	A	/	H		A
Team 3	H	A	/		H
				/	
Team n	A	H	A		/

We want to measure how much a given schedule deviates from being ranking fair. Clearly, an individual ranking HAP of the form HAH...HAA, while not ranking fair, is better than HH...HAA...A

A new concept: ranking fairness

We want to measure how much a given schedule deviates from being ranking fair. Clearly, an individual ranking HAP of the form HAHA ...HAA, while not ranking fair, is better than HH...HAA...A

For team p , we consider all consecutive intervals of the ranking HAP with left and right indices $1 \leq i < j \leq n-1$. Let H_{ij}^p denote the number of home matches of team p that occur in this sub-pattern.

Given a schedule we define the *ranking fairness* of a team p as:

$$F_p = \sum_{i=1}^{n-1} \sum_{j=i+1}^n \left| H_{ij}^p - \frac{j-i+1}{2} \right| - \frac{(n-2)^2}{8}$$

The *ranking fairness* of a schedule is now defined as: $F = \sum_p F_p$.

Claim: $F = 0$ is equivalent to a schedule being ranking fair.

The scheduling problem

- Do there exist single-break, ranking fair schedules?

The scheduling problem

□ Do there exist single-break, ranking fair schedules?

□ There is one for $n=4$:

R1 R2 R3
1 vs 2 3 vs 1 1 vs 4
3 vs 4 4 vs 2 2 vs 3

	1	2	3	4
1	\	H	A	H
2		\	H	A
3			\	H
4				\

The scheduling problem

□ Do there exist single-break, ranking fair schedules?

□ There is one for $n=4$:

R1	R2	R3
1 vs 2	3 vs 1	1 vs 4
3 vs 4	4 vs 2	2 vs 3

	1	2	3	4
1	\	H	A	H
2		\	H	A
3			\	H
4				\

The combined view:

	1	2	3	4
1	\	1	@2	3
2	@1	\	3	@2
3	2	@3	\	1
4	@3	2	@1	\

A cell (i,j) contains the round where teams i and j meet

The scheduling problem

- Do there exist single-break, ranking fair schedules?
- There is none for $n=6$.

The scheduling problem

□ Do there exist single-break, ranking fair schedules?

□ There is one for $n=8$:

R1	R2	R3	R4	R5	R6	R7
1 vs 4	3 vs 1	1 vs 2	5 vs 1	1 vs 6	7 vs 1	1 vs 8
7 vs 3	2 vs 5	3 vs 4	2 vs 3	3 vs 8	5 vs 3	3 vs 6
5 vs 8	4 vs 7	5 vs 6	6 vs 7	7 vs 5	6 vs 4	4 vs 5
6 vs 2	8 vs 6	7 vs 8	8 vs 4	4 vs 2	8 vs 2	2 vs 7

	1	2	3	4	5	6	7	8
1	\	H	A	H	A	H	A	H
2		\	H	A	H	A	H	A
3			\	H	A	H	A	H
4				\	H	A	H	A
5					\	H	A	H
6						\	H	A
7							\	H
8								\

The scheduling problem

- Do there exist single-break, ranking fair schedules?

	1	2	3	4	5	6	7	8
1	\	7	@6	5	@4	3	@2	1
2	@7	\	5	@6	3	@4	1	@2
3	6	@5	\	3	@2	1	@7	4
4	@5	6	@3	\	1	@2	4	@7
5	4	@3	2	@1	\	7	@5	6
6	@3	4	@1	2	@7	\	6	@5
7	2	@1	7	@4	5	@6	\	3
8	@1	2	@4	7	@6	5	@3	\

The combined view

The scheduling problem

- Do there exist single-break, ranking fair schedules?
- There is none for $n=10$.

The scheduling problem

- Do there exist single-break, ranking fair schedules?
- There is none for $n=10$.
- There is one for $n=12$.

The scheduling problem

□ Do there exist single-break, ranking fair schedules?

	1	2	3	4	5	6	7	8	9	10	11	12
1	\	11	@10	9	@8	7	@6	5	@4	3	@2	1
2	@7	\	9	@10	7	@8	5	@6	3	@4	1	@2
3	10	@9	\	7	@6	5	@4	3	@2	1	@11	8
4	@9	10	@7	\	5	@6	3	@4	1	@2	8	@11
5	8	@7	6	@5	\	3	@2	1	@10	11	@9	4
6	@7	8	@5	6	@3	\	1	@2	11	@10	4	@9
7	6	@5	4	@3	2	@1	\	11	@9	8	@7	10
8	@5	6	@3	4	@1	2	@11	\	8	@9	10	@7
9	4	@3	2	@1	11	@10	9	@8	\	7	@5	6
10	@3	4	@1	2	@10	11	@8	9	@7	\	6	@5
11	2	@1	11	@8	9	@4	7	@10	5	@6	\	3
12	@1	2	@8	11	@4	9	@10	7	@6	5	@3	\

The scheduling problem

- Do there exist single-break, ranking fair schedules?
- There is none for $n=14$.

The scheduling problem

□ Summarizing

- $n=4$: yes
- $n=6$: no
- $n=8$: yes
- $n=10$: no
- $n=12$: yes
- $n=14$: no
- $n=16$: yes
- $n=18$: ?

The scheduling problem

□ Summarizing

- $n=4$: yes
- $n=6$: no
- $n=8$: yes
- $n=10$: no
- $n=12$: yes
- $n=14$: no
- $n=16$: yes
- $n=18$: yes!

Results for the scheduling problem

Theorem: for $n=4k$, there exists a single break, ranking fair schedule

The corresponding D-sequence is $313131\dots312$

Results for the scheduling problem

Theorem: for $n=4k$, there exists a single break, ranking fair schedule

The corresponding D-sequence is 313131...312

Observation: for every value of $n=4k+2$ where $18 \leq n \leq 102$ single break, ranking fair schedules exist.

Open question

What about arbitrary values of n of the form $n=4k+2$?

Our results

Theorem: the Canonical Pattern Set (CPS) does not allow a ranking fair schedule for $n \geq 12$.

Proof. Consider the HAPS from the CPS with $|H| = |A| + 1$. We call these patterns p_r , where index r ($1 \leq r \leq n-1$) refers to the round where the H break of pattern p_r is. Let $t(p_r)$ refer to the team assigned to pattern p_r .

Identify the r in $\{1, \dots, n-1\}$ with $t(p_r) = 1$. As we deal with the CPS we have patterns with breaks in rounds $r+2, r+4, r+6$ or $r-2, r-4, r-6$. Assume wlog the former.

r-1	r	r+1	r+2	r+3	r+4	r+5	r+6
H	H	A	H	A	H	A	H
H	A	H	H	A	H	A	H
H	A	H	A	H	H	A	H
H	A	H	A	H	A	H	H

Our results

Theorem: the Canonical Pattern Set (CPS) does not allow a ranking fair schedule for $n \geq 12$.

Proof. Consider the HAPS from the CPS with $|H| = |A| + 1$. We call these patterns p_r , where index r ($1 \leq r \leq n-1$) refers to the round where the H break of pattern p_r is. Let $t(p_r)$ refer to the team assigned to pattern p_r .

Identify the r in $\{1, \dots, n-1\}$ with $t(p_r) = 1$. As we deal with the CPS we have patterns with breaks in rounds $r+2, r+4, r+6$ or $r-2, r-4, r-6$. Assume wlog the former.

r-1	r	r+1	r+2	r+3	r+4	r+5	r+6
H	H	A	H	A	H	A	H
H	A	H	H	A	H	A	H
H	A	H	A	H	H	A	H
H	A	H	A	H	A	H	H

Our results

Theorem: the Canonical Pattern Set (CPS) does not allow a ranking fair schedule for $n \geq 12$.

Proof. Consider the HAPS from the CPS with $|H| = |A| + 1$. We call these patterns p_r , where index r ($1 \leq r \leq n-1$) refers to the round where the H break of pattern p_r is. Let $t(p_r)$ refer to the team assigned to pattern p_r .

Identify the r in $\{1, \dots, n-1\}$ with $t(p_r) = 1$. As we deal with the CPS we have patterns with breaks in rounds $r+2, r+4, r+6$ or $r-2, r-4, r-6$. Assume wlog the former.

r-1	r	r+1	r+2	r+3	r+4	r+5	r+6
H	H	A	H	A	H	A	H
H	A	H	H	A	H	A	H
H	A	H	A	H	H	A	H
H	A	H	A	H	A	H	H

Incomplete Round Robin Tournaments

A motivating story in football

The Champions League since 2024-2025

- There are 36 teams, partitioned into four, strength-based, sets (pots) of 9 teams each.
- Each team plays 8 matches (4 home, 4 away) against 8 different teams, 2 from each pot. A win gives the team 3 points, a draw 1 point, and a loss 0.
- Two important constraints: (i) no team plays against a team from the same domestic league, (ii) no team plays against more than 2 teams from the same domestic league.
- After the 144 matches, the teams are ranked based on the number of collected points (and a series of tiebreakers).

A motivating story in football

The Champions League since 2024-2025

- ❑ This format is a so-called *incomplete* round robin.
- ❑ A key concern: since not every team plays the same set of opponents, is this a fair tournament? More specifically, will the resulting ranking be a true representation of the strength of the teams??
- ❑ UEFA arrived at the 36 opponent sets by using a draw whose outcome is guaranteed to satisfy the constraints mentioned above.

A motivating story in football

The Champions League since 2024-2025

- ❑ Fact: the 36 teams vary quite a bit in strength.
- ❑ Let us associate a number to each team indicating its strength. I use the Elo rating.



UEFA
CHAMPIONS
LEAGUE

A motivating story in football

	Elo		Elo
Man City	2060	VfB Stuttgart	1791
Real M	1988	Aston Villa	1778
Inter Milan	1966	Lille	1771
Arsenal	1950	Monaco	1770
Liverpool	1918	Bologna	1769
Bayer Leverkusen	1918	Benfica	1759
Bayern Muenchen	1908	Feyenoord	1748
Barcelona	1898	Sparta Praag	1728
PSG	1895	Club Brugge	1709
Borussia D	1870	Brest	1685
Atalanta	1866	Red Bull Salzburg	1674
RB Leipzig	1861	Celtic	1653
Atletico M	1838	Sturm Graz	1606
Sporting CP	1835	Dinamo Zagreb	1583
Juventus	1833	Shaktar Donetsk	1576
AC Milan	1818	Rode Ster Belgrado	1568
Girona	1801	Young Boys	1553
PSV	1797	Slovan Bratislava	1446

The 36 teams and their Elo ratings in 2024-2025

A motivating story in football



	Elo		Elo
Man City	2060	VfB Stuttgart	1791
Real M	1988	Aston Villa	1778
Inter Milan	1966	Lille	1771
Arsenal	1950	Monaco	1770
Liverpool	1918	Bologna	1769
Bayer Leverkusen	1918	Benfica	1759
Bayern Muenchen	1908	Feyenoord	1748
Barcelona	1898	Sparta Praag	1728
PSG	1895	Club Brugge	1709
Borussia D	1870	Brest	1685
Atalanta	1866	Red Bull Salzburg	1674
RB Leipzig	1861	Celtic	1653
Atletico M	1838	Sturm Graz	1606
Sporting CP	1835	Dinamo Zagreb	1583
Juventus	1833	Shaktar Donetsk	1576
AC Milan	1818	Rode Ster Belgrado	1568
Girona	1801	Young Boys	1553
PSV	1797	Slovan Bratislava	1446

Average Elo:
 $\bar{s} \approx 1783.89$

A motivating story in football



	Elo	Average opponent Elo		Elo	Average opponent Elo
PSG	1895	1853	Arsenal	1950	1786
Liverpool	1918	1840	Bologna	1769	1785
Sparta Praag	1728	1835	Monaco	1770	1784
Feyenoord	1748	1827	Brest	1685	1773
Lille	1771	1817	Girona	1801	1772
Juventus	1833	1816	Man City	2060	1770
Sporting CP	1835	1815	PSV	1797	1764
Club Brugge	1709	1807	Young Boys	1553	1762
Slovan Bratislava	1446	1807	Dinamo Zagreb	1583	1761
RB Leipzig	1861	1804	Aston Villa	1778	1757
Inter Milan	1966	1801	Atletico M	1838	1756
Sturm Graz	1606	1800	Atalanta	1866	1752
Shaktar Donetsk	1576	1800	Barcelona	1898	1747
Real M	1988	1799	AC Milan	1818	1741
Red Bull Salzburg	1674	1798	VfB Stuttgart	1791	1735
Bayer Leverkusen	1918	1797	Borussia D	1870	1723
Rode Ster Belgrado	1568	1794	Bayern Muenchen	1908	1710
Benfica	1759	1792	Celtic	1653	1708

The 36 teams,
their Elo
ratings, and
their
opponent sets
Elo ratings
in 2024-2025

A motivating story in football

- Some observations.

- Celtic had the weakest set of opponents: 1708

{Bor D, Atalanta, Leipzig, Club Brugge, Dinamo, Young Boys, Aston Villa} – Celtic ended up no 18

- PSG had the strongest set of opponents: 1853

{Man City, Atletico M, PSV, Girona, Bayern, Arsenal, RB Salzburg, VfB Stuttgart} – PSG was ranked 22th after 7 rounds

- But these observations relevant for fans; we should look at an aggregate measure, say the max – min; let us call that the *bandwidth*.

A motivating story in football



	Elo	Average opponent Elo
PSG		1849
PSV		1828
Union		1827
Newcastle		1817
Leverkusen		1817
Frankfurt		1812
Qarabag		1811
Athletic Club		1811
Marseille		1810
Slavia Praha		1810
Atleti		1809
Benfica		1800
B Dortmund		1799
Bayern Munich		1796
Manchester City		1796
Bodo/Glimt		1794
Barcelona		1789
Club Brugge		1789

The 36 teams,
their Elo
ratings, and
their
opponent sets
Elo ratings
in 2025-2026

	Elo	Average opponent Elo
Kairat Almaty		1786
Atalanta		1783
Galasataray		1781
Liverpool		1779
Napoli		1776
Sporting CP		1772
Monaco		1767
Ajax		1765
Tottenham		1764
Juventus		1764
Chelsea		1761
Real Madrid		1761
Villareal		1756
Olympiacos		1749
Inter		1741
Arsenal		1740
Pafos		1740
Copenhagen		1733

A motivating story in football

Does a fairer draw for the Champions League exist?

Season	Actual bandwidth	Achievable bandwidth
2024-2025	145	?
2025-2026	116	?

Champions League: a fair draw?

Here is a mathematical model for a fairest draw.

- Let P be the set of teams, partitioned into P_ℓ denoting the set of teams in pot $\ell=1,2,3,4$. Let s_j be the Elo rating of team j in P .
- We have, for each pair of teams i,j , a binary variable $x_{i,j}$ which equals 1 if and only if team j is in an 8-tuple selected for team i . We use two real variables z_{max} and z_{min} that capture the bandwidth between the strongest and weakest opponent set.

Champions League: a fair draw?

minimize $z_{max} - z_{min}$

s.t. $\sum_{j \in P_\ell} x_{ij} = 2$

for $i \in P, \ell=1,2,3,4$

$x_{ij} - x_{ji} = 0$

for distinct $i, j \in P$

$z_{max} - \sum_{j \in P} s_j x_{ij} / 8 \geq 0$

for $i \in P$

$z_{min} - \sum_{j \in P} s_j x_{ij} / 8 \leq 0$

for $i \in P$

$x_{ij} \in \{0,1\}$

for distinct $i, j \in P$

A motivating story in football

Does a fairer draw for the Champions League exist?

Season	Actual bandwidth	Achievable bandwidth
2024-2025	145	?
2025-2026	116	?

A fairer draw for the Champions League

Season	Actual bandwidth	Achievable bandwidth
2024-2025	145	1.653
2025-2026	116	3.250

A motivating story in football

The draw might produce opponent sets that are very different in strength, or in other words, unfair.

Towards the scheduling problem: terminology

A *draw* is a set of $nk/2$ matches (unordered pairs of teams) such that each team occurs in k matches.

The *opponent set* of a team i is the set of k opponents team i plays against in a draw.

The *strength* of an opponent set is the average strength of the teams in the opponent set.

The *bandwidth* equals the difference between the maximum strength of an opponent set and the minimum strength of an opponent set.

The scheduling problem

Given k (number of matches), and given n (number of teams) with their strengths s_i , does there exist a fair draw?

In other words, does there exist a draw with bandwidth 0?

In other words, does there exist a draw where each opponent set has strength $\bar{s} = \frac{\sum_i s_i}{n}$?

The problem for $k=2$

Let's build a graph $G=(V,E)$ where

- V corresponds to the set of teams, and
- $E = \{ \{i,j\} : \frac{s_i + s_j}{2} = \bar{s} \}$
- Notice that E contains the allowed opponent sets.

The problem for $k=2$

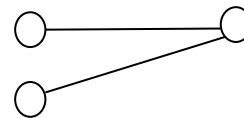
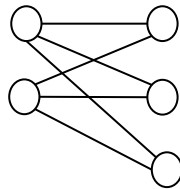
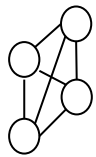
□ Two questions:

1. How does G look like??
2. What does a fair draw look like?

The problem for $k=2$

1. How does G look like??

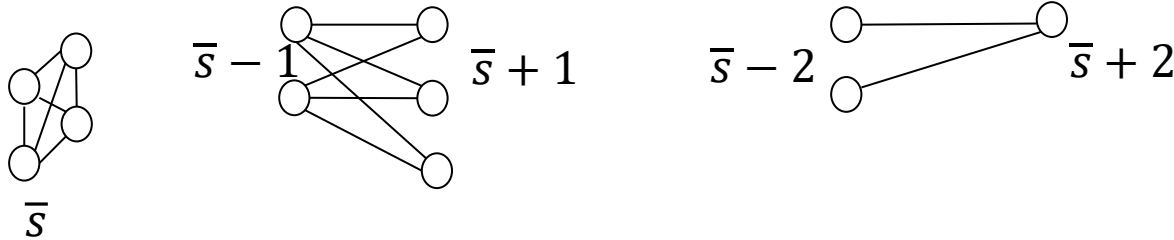
G is a union of a clique and complete bipartite subgraphs



The problem for $k=2$

1. How does G look like??

G is a union of a clique and complete bipartite subgraphs



The problem for $k=2$

2: What does a fair draw look like?

The problem for $k=2$

2: What does a fair draw look like?

Answer: A fair draw is a *perfect 2-matching* in G

A perfect 2-matching is an assignment of weights $w(e) \in \{0,1,2\}$ for every $e \in E$ such that $\sum_{e \in \delta(v)} w(e) = 2$ for each $v \in V$.

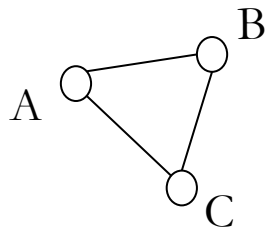
The problem for $k=2$

2: What does a fair draw look like?

Answer: A fair draw is a *perfect 2-matching* in G

A perfect 2-matching is an assignment of weights $w(e)$ in $\{0,1,2\}$ for every $e \in E$ such that $\sum_{e \in \delta(v)} w(e) = 2$ for each $v \in V$

Example



Draw:

A: {B,C}

B: {A,C}

C: {A,B}

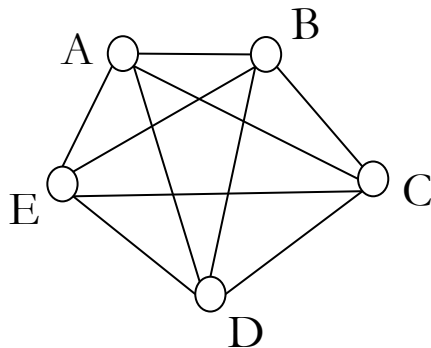
The problem for $k=2$

2: What does a fair draw look like?

Answer: A fair draw is a *perfect 2-matching* in G

A perfect 2-matching is an assignment of weights $w(e)$ in $\{0,1,2\}$ for every $e \in E$ such that $\sum_{e \in \delta(v)} w(e) = 2$ for each $v \in V$.

Example



Draw:

A: {C,D}

B: {D,E}

C: {A,E}

D: {A,B}

E: {B,C}

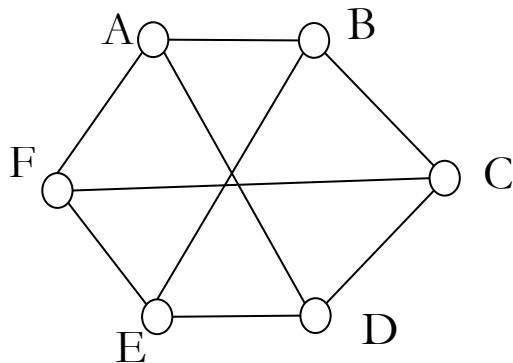
The problem for $k=2$

2: What does a fair draw look like?

Answer: A fair draw is a *perfect 2-matching* in G

A perfect 2-matching is an assignment of weights $w(e) \in \{0,1,2\}$ for every $e \in E$ such that $\sum_{e \in \delta(v)} w(e) = 2$ for each $v \in V$.

Another example



Draw:

There is none
while there is a perfect 2-matching

The problem for $k=2$

□ Let n_0 denote the number of teams with strength $\bar{s}$

Theorem: There exists a fair draw for iRR with $k=2$ precisely when every bipartite component of G has a perfect matching and either:

- $n_0 = 0, 2, 4$ and n is a multiple of 4, or
- $n_0 = 3$ and $n-3$ is a multiple of 4, or
- $n_0 \geq 5$.

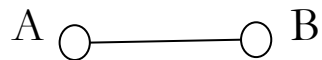
The problem for $k=2$

□ Let n_0 denote the number of teams with strength $\bar{s}$

Theorem: There exists a fair draw for iRR with $k=2$ precisely when every bipartite component of G has a perfect matching and either:

- $n_0 = 0, 2, 4$ and n is a multiple of 4, or
- $n_0 = 3$ and $n-3$ is a multiple of 4, or
- $n_0 \geq 5$.

Basic idea:



Matches: A: {C,D}

B: {C,D}

C: {A,B}

D: {A,B}

The problem for $k=2$

□ Let n_0 denote the number of teams with strength $\bar{s}$

Theorem: There exists a fair draw for iRR with $k=2$ precisely when every bipartite component of G has a perfect matching and either:

- $n_0 = 0, 2, 4$ and n is a multiple of 4, or
- $n_0 = 3$ and $n-3$ is a multiple of 4, or
- $n_0 \geq 5$.

Corollary: iRR for $k=2$ can be solved in polynomial time.

About connectedness of the draw

Given the draw, we can construct the *draw graph*, $H=(V,E)$, where V represents the set of teams, and E represents the set of matches.

We call a draw *connected* when the draw graph is connected.

About connectedness of the draw

Given the draw, we can construct the *draw graph*, $H=(V,E)$, where V represents the set of teams, and E represents the set of matches.

We call a draw *connected* when the draw graph is connected.

Theorem: there exists a fair connected draw for iRR for $k = 2$, precisely when a fair draw exists and either:

- ▣ all teams have the same strength, or
- ▣ $n/2$ teams have strength $s - d$ and $s + d$, respectively, or
- ▣ $n/2$ teams have strength s , $n/4$ teams have strength $s - d$ and $s + d$, or
- ▣ $n/4$ teams have strengths $s - d, s + d, s - e, s + e$.

The problem for $k=3$

Theorem: iRR is NP-complete for $k = 3$.

We use a reduction from Numerical 3-Dimensional Matching.

Computational Experiments

How do random draws differ from minimum-bandwidth draws?

We generate instances with $n \in \{16, 32\}$, $k \in \{4, 8, 12\}$. Strengths are from a uniform distribution $[0, 100]$.

Computational Experiments

Let's look at bandwidth (each entry is the average of a 100 instances).

n=16	Average Bandwidth of Random draw	Minimum Bandwidth
k=4	45.25	0.38
k=8	26.84	0.06
k=12	14.70	0.13

n=32	Average Bandwidth of Random draw	Minimum Bandwidth
k=4	55.24	1.23
k=8	36.25	0.41
k=12	27.76	0.23

Computational Experiments

Let's look at connectedness of the draw.

n=16	Diameter of the draw graph from a random draw	Average diameter of the draw graph of a minimum-bandwidth draw
k=4	3.19	4.00
k=8	2.00	2.00
k=12	2.00	2.00

n=32	Diameter of the draw graph from a random draw	Average diameter of the draw graph of a minimum-bandwidth draw
k=4	4.35	4.70
k=8	3.00	3.00
k=12	2.11	2.00

Finding

- ▣ Minimum bandwidth draws have significant smaller bandwidth than random draws, and are a little bit less connected.

Conclusion

- ❑ Scheduling in Sports: there is a world to conquer!

References

- Wessel, S., C. Hurkens, and F.C.R. Spieksma (2026), *Fair Schedules for Single Round Robin Tournaments with Ranked Participants*, European Journal of Operational Research **335**, 829-838.
- Wessel, S. and F.C.R. Spieksma (2026), *Finding Fair Draws For Incomplete Round Robin Tournaments*, ArXiv: 2607.17901.

The end

Thanks!